

RESONANCE-MARKED NATURALITY AND HOMOTOPY-TILT NON-INVARIANCE IN THE INVERSE-LEIBNIZ CUBIC CASE

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ABSTRACT. We determine which features of a fixed cubic inverse-Leibniz calculation survive changes of its normalized homotopy splitting. In a seven-dimensional recurrent envelope, the full internal tilt orbit consists of all operators satisfying $A(E) = -2E$. Its common characteristic and minimal polynomial divisor is $S + 2$; the complementary quintic factor of the chosen contraction can be removed. We then define a primitive-free resonant boundary-response line and a response covector on the tangent subspace for which the bracket descends through the primitive quotient. We characterize this domain as the kernel of an explicit descent obstruction; any output quotient extending a transverse cubic direction must erase the marked line. The response is transported by strict comparisons preserving the marked line, differential, and bracket. An explicit four-generator DGLA has a contractible resonant pair that a strict quasi-isomorphism cancels, while the Maurer–Cartan functor over local Artin algebras remains a smooth two-parameter germ. Thus the positive comparison theorem requires the resonance marking, which the ordinary deformation functor does not retain. All spectral claims concern the stated internal orbit; the finite cubic input and the available computational checks are identified explicitly.

1. SCOPE AND MAIN CONCLUSIONS

The inverse-Leibniz cubic calculation produces two kinds of output. The first kind belongs to a chosen contraction: a controller box, a recurrent matrix, a cyclic source generator, and a polynomial such as

$$m_A(S) = (S + 2)q(S).$$

The second kind is forced by the original cochain identities together with a normalized boundary direction. The purpose of this paper is to separate these two layers and to identify the correct comparison statement.

The main conclusions are:

- (i) the exact recurrent operator is not invariant under admissible homotopy tilts;
- (ii) the normalized E -rail and its -2 response are invariant inside the fixed envelope;
- (iii) the primitive-free resonant boundary-response pair is natural under strict resonance-faithful comparisons;
- (iv) ordinary DGLA quasi-isomorphism does not preserve that pair, even when the formal Maurer–Cartan deformation behavior is unchanged.

The paper therefore ends with the boxed scope statement

strict/resonance-faithful naturality + ordinary quasi-isomorphism no-go.

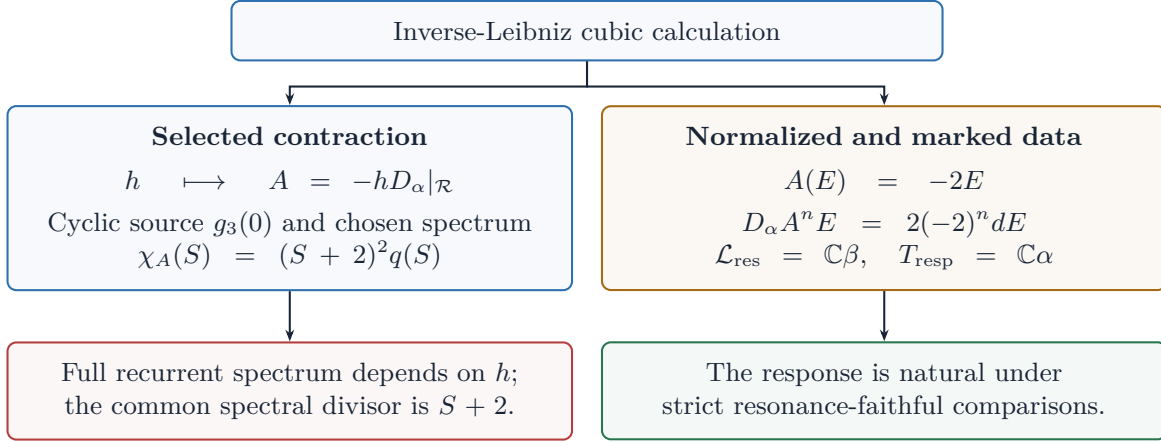
No claim is made here for arbitrary triangular systems with spectral multiplicity vector

$$m_A(S) = \prod_{\lambda} (S - \lambda)^{e_{\lambda}}.$$

Relation to existing deformation theory. The fixed action-reconstruction problem and its selected contraction are developed in Paper I [4]. The deformation functors used for comparison are standard Maurer–Cartan constructions [1, 2]. Homotopy transfer changes representatives and higher operations [3, Chapter 10]; it does not by itself assert invariance of the full matrix $-hD_\alpha$. Here the additional information is a specified boundary line and its response. The explicit cancellation example below proves the failure of forgetting this mark without appealing to a general equivalence theorem.

Paper III [5] extends the polynomial-floor calculation to finite marked presentations. The present cubic result supplies its motivating example. The two complete contractions in the v0.08 Paper I comparison agree on the saved low-rail data; they illustrate finite higher-transfer dependence, not an identification of those two completions with every parameter in the internal tilt orbit.

Figure 1 summarizes the two kinds of data and the comparison statements that separate them.



Normalized internal tilts keep the E -rail in the fixed envelope.
Ordinary quasi-isomorphism can erase the marked boundary-response pair.

FIGURE 1. Contraction-dependent and marked data. The common divisor $S + 2$ concerns normalized internal tilts in the fixed envelope. Naturality of the primitive-free response additionally requires a resonance-faithful comparison; ordinary quasi-isomorphism need not retain the marked boundary line.

2. THE FIXED CUBIC RECURRENT ENVELOPE

Throughout this paper, α, ξ, E denote the Hochschild cochains $\mathbf{a}, \mathbf{x}, \mathbf{e}$ of Paper I, not the axis perturbations before their Hochschild lift. The symbol E here is a cochain, not the polynomial $S + 2$. The differential and cup-commutator bracket are those of Paper I's constrained Hochschild DGLA. Its subsection *The presentation quartic class does not lift intrinsically* and its low-order source appendix specify $\mathbf{k}, \mathbf{p}$. For later use, the source dictionary is

$$g_3(k) = x_3(k+1) + 2x_3(k), \quad g_3(0) = 2\mathbf{k} - \mathbf{p},$$

$$Y_{13} = [\alpha, \mathbf{k}] - 2[\xi, E], \quad C_{13} = Y_{13} - d\mathbf{p}.$$

Thus C_{13} is a nonclosed complement vector, while dE is a nonzero boundary. The ordered cochain data and exact replay entry points are identified in the companion's `EVIDENCE_MAP.md`.

Let

$$C^1 \xrightarrow{d} C^2 \xrightarrow{d} C^3$$

be the relevant constrained Hochschild complex, and let

$$D_\alpha = [\alpha, -] : C^1 \longrightarrow C^2$$

be the operator determined by the cocycle α tangent to the reduced branch. A selected partial contraction supplies a finite-dimensional recurrent envelope

$$\mathcal{R} \subset C^1, \quad A = -hD_\alpha|_{\mathcal{R}} : \mathcal{R} \longrightarrow \mathcal{R}.$$

More explicitly, for this selected homotopy the ordered recurrent basis is

$$(E, g_3(0), Ag_3(0), \dots, A^5 g_3(0)).$$

These seven cochains are independent and span the invariant envelope $\mathcal{R} = \mathbb{C}E \oplus \mathbb{C}[A]g_3(0)$. The six-dimensional cyclic summand and its companion matrix are established in Paper I's section *Triangular spectral extension and residual modules*. The exact cubic records [6] verify the extra E direction and the source intersections, giving

$$\dim \mathcal{R} = 7, \quad \text{rank}(D_\alpha|_{\mathcal{R}}) = 7,$$

and

$$D_\alpha(\mathcal{R}) \cap Z^2 = D_\alpha(\mathcal{R}) \cap B^2 = \mathbb{C}D_\alpha E.$$

The normalized direction satisfies

$$D_\alpha E = 2dE = -2\xi \smile \xi, \quad A(E) = -2E.$$

For the fixed contraction, the characteristic polynomial is

$$\chi_A(S) = (S + 2)^2 q(S), \quad q(S) = S^5 - 4S^4 + 12S^3 - 32S^2 + 80S - 192.$$

This formula is exact, but its status is deliberately local: it describes the selected partial contraction before quotienting by homotopy tilts.

The selected envelope and its source image are displayed in Figure 2.

Proposition 2.1 (Boundary-compatible completion of the finite source data). *Let $U = U_{16}$, let $K_{\text{old}} \subset C^2$ be the fixed 100-dimensional provenance source space, let $F = D_\alpha(U)$, and put $L = K_{\text{old}} + F$. The reconstructed exact data satisfy*

$$\dim U = \text{rank}(d|_U) = 35, \quad \dim F = 23, \quad \dim L = 116,$$

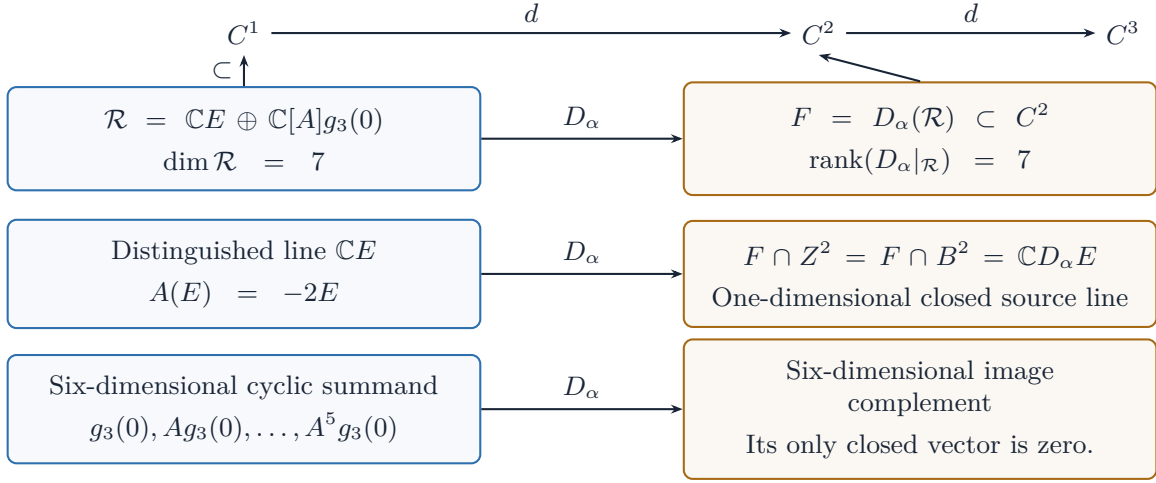
$$\dim(F \cap K_{\text{old}}) = 7, \quad L \cap B^2 = K_{\text{old}} \cap B^2 = \mathbb{C}dE.$$

The fixed map $h_{\text{old}} : K_{\text{old}} \rightarrow U$ has $h_{\text{old}}(dE) = E$. Its extensions to L , with values in U , form an affine space modeled on

$$\text{Hom}(L/K_{\text{old}}, U),$$

of dimension $16 \cdot 35 = 560$. Each has a boundary-compatible extension to a standard contraction, allowing the remaining splitting choices to vary.

Proof. The ranks and intersections are recomputed from the cochains by the v0.03 supplement; the full constrained boundary space has dimension 88. A linear extension is free on a complement of K_{old} in L . Since $d|_U$ is injective and $L \cap dU = \mathbb{C}dE$, every such extension agrees on that intersection with $(d|_U)^{-1} : dU \rightarrow U$. It therefore extends consistently to $L + dU$ with $hd|_U = \text{id}$. Choose the degree-one acyclic complement to contain U , extend the inverse on boundaries, and choose cycle representatives inside the resulting kernel of h . On a closed



$$D_\alpha = [\alpha, -], \quad D_\alpha E = 2dE = -2\xi \smile \xi.$$

All decompositions here refer to the selected partial contraction.

FIGURE 2. The fixed seven-dimensional recurrent envelope. The ordered basis is $(E, g_3(0), Ag_3(0), \dots, A^5 g_3(0))$. The restriction of D_α is injective, and its closed source intersection is exactly the normalized boundary line. This figure does not assert invariance of the envelope under arbitrary changes of contraction.

element $z \in L$, the vector $z - dhz$ is closed; the one-dimensional boundary intersection ensures the required cycle representatives are independent modulo boundaries. Extending these vector-space splittings degree by degree gives a standard contraction. No previously unspecified cohomology representatives are held fixed in this assertion. $\square$

Remark 2.2. *Cycles must not be confused with boundaries here: the recomputed values are $\dim(L \cap Z^2) = 2$ and $\dim(L \cap B^2) = 1$. The additional cohomology direction is compatible with the splitting construction; it cannot be discarded when checking contraction identities.*

3. THE EXACT HOMOTOPY-TILT ACTION

Definition 3.1 (Normalized source directions). *Put*

$$F = D_\alpha(\mathcal{R}), \quad F_{\text{norm}} = F \cap Z^2, \quad \overline{F} = F/F_{\text{norm}}.$$

The normalized internal tilt group is

$$\mathbb{T}_{\mathcal{R}} = \text{Hom}(\overline{F}, \mathcal{R}).$$

For $\tau \in \mathbb{T}_{\mathcal{R}}$, write $q_F : F \rightarrow \overline{F}$ for the quotient map and define

$$h_\tau(q) = h(q) + \tau(q_F(q)), \quad q \in F.$$

The normalized directions are fixed because q_F vanishes on F_{norm} . The residual source changes by

$$r_\tau(q) = q - dh_\tau(q), \quad dr_\tau(q) = dq.$$

Thus the degree-three differential data of the nonclosed source quotient are unchanged.

Proposition 3.2 (Exact operator action). *The recurrent operator changes according to*

$$A_\tau = A - \tau q_F D_\alpha|_{\mathcal{R}}.$$

The map

$$\mathbb{T}_{\mathcal{R}} \longrightarrow \{T \in \text{End}(\mathcal{R}) : T(E) = 0\}, \quad \tau \longmapsto \tau q_F D_\alpha|_{\mathcal{R}},$$

is an isomorphism in the cubic model.

Proof. The first identity is immediate from $A = -hD_\alpha$ and the definition of h_τ . Since $D_\alpha|_{\mathcal{R}}$ has rank seven while $F_{\text{norm}} = \mathbb{C}D_\alpha E$, the quotient map $q_F D_\alpha$ has kernel $\mathbb{C}E$ and identifies

$$\mathcal{R}/\mathbb{C}E \cong \overline{F}.$$

Every endomorphism vanishing on E therefore factors uniquely through $q_F D_\alpha$. $\square$

Lemma 3.3 (The internal tilts are realizable splittings). *Every tilt in the displayed internal family extends to a contraction of the full constrained complex. Only its stated normalization is fixed; the family does not require retention of all 100 archived homotopy assignments.*

Proof. The recurrent envelope lies in the 35-dimensional space U on which d is injective. Choose a degree-one acyclic complement containing U . The variation τq_F on F vanishes on $F \cap Z^2$ and has image in that complement. Extend it by zero on Z^2 and then linearly to C^2 . Adding it to a completed homotopy preserves its inverse on boundaries and its zero values on the chosen cohomology representatives. The resulting map h' has image in the same acyclic complement and is surjective onto it. Inside $\ker h'$, extend the chosen cohomology representatives by a complement W ; because $\ker h' \cap Z^2$ is exactly those representatives, W is a complement to Z^2 . Projection along $B^2 \oplus W$ and the adjacent-degree splittings give the required contraction. $\square$

Corollary 3.4 (Cubic tilt orbit). *In the fixed seven-dimensional envelope,*

$$\dim \mathbb{T}_{\mathcal{R}} = 6 \cdot 7 = 42,$$

and

$$\mathcal{O}_{\text{tilt}}(A) = \{B \in \text{End}(\mathcal{R}) : B(E) = -2E\}.$$

Proof. The dimension follows from $\dim \overline{F} = 6$ and $\dim \mathcal{R} = 7$. The preceding proposition says precisely that all variations vanishing on E occur, while the normalized relation $A(E) = -2E$ is fixed. $\square$

The number 42 here is the dimension of a parameter space $\text{Hom}(\mathcal{R}/\mathbb{C}E, \mathcal{R})$. It is not the full cohomology space H^2 of Paper I, whose separately computed dimension happens also to be 42. Nor is this 42-parameter family the 560-parameter extension family fixing all of K_{old} in Proposition 2.1.

4. SPECTRAL NON-INVARIANCE AND THE SURVIVING RAIL

The tilt orbit contains the two exact representatives

$$B_0 = \text{diag}(-2, 0, 0, 0, 0, 0, 0), \quad B_3 = \text{diag}(-2, 3, 3, 3, 3, 3, 3).$$

Their characteristic polynomials are

$$\chi_{B_0}(S) = (S + 2)S^6, \quad \chi_{B_3}(S) = (S + 2)(S - 3)^6.$$

Both operators agree with A on $\mathbb{C}E$ and hence are exact normalized internal tilts. We obtain:

Theorem 4.1 (Homotopy-tilt spectral no-go). *The factor $q(S)$, the second -2 eigenline, the six-dimensional cyclic module generated by the fixed $g_3(0)$, and the fixed-contraction recurrence $(S+2)q(S)$ are not invariants of the original cubic deformation problem under normalized internal homotopy tilts. The greatest common divisor of the characteristic polynomials over the full internal tilt orbit is*

$$\gcd_{\tau} \chi_{A_{\tau}}(S) = S + 2,$$

and the same statement holds for minimal polynomials.

Proof. The two displayed representatives already change the complementary spectrum. Conversely every operator in the orbit has the eigenrelation $B(E) = -2E$, so $S + 2$ divides every characteristic and minimal polynomial. Since the induced operator on $\mathcal{R}/\mathbb{C}E$ is arbitrary, no nonconstant factor on that quotient can be common to the whole orbit. $\square$

Figure 3 makes the full internal orbit explicit. Relative to any chosen vector-space complement W of $\mathbb{C}E$, the off-diagonal component $W \rightarrow \mathbb{C}E$ and the quotient operator are both arbitrary; the complement need not be invariant.

$$\begin{aligned} \mathbb{T}_{\mathcal{R}} &= \text{Hom}(F/F_{\text{norm}}, \mathcal{R}), \quad \dim \mathbb{T}_{\mathcal{R}} = 42, \quad A_{\tau} = A - \tau q_F D_{\alpha}|_{\mathcal{R}} \\ \mathcal{O}_{\text{tilt}}(A) &= \{B \in \text{End}(\mathcal{R}) : B(E) = -2E\} \end{aligned}$$

Choose a vector-space complement W : $\mathcal{R} = \mathbb{C}E \oplus W$, $W \cong \mathcal{R}/\mathbb{C}E$.

$$B = \begin{pmatrix} -2 & \ell \\ 0 & C \end{pmatrix}, \quad \ell \in \text{Hom}(W, \mathbb{C}E), \quad C \in \text{End}(W) \text{ arbitrary.}$$

The chosen complement need not be B -invariant.

$$\begin{aligned} \text{Selected } A \\ \chi_A(S) &= (S+2)^2 q(S) \end{aligned}$$

$$\begin{aligned} B_0 &= \text{diag}(-2, 0, \dots, 0) \\ \chi_{B_0}(S) &= (S+2)S^6 \end{aligned}$$

$$\begin{aligned} B_3 &= \text{diag}(-2, 3, \dots, 3) \\ \chi_{B_3}(S) &= (S+2)(S-3)^6 \end{aligned}$$

$$\gcd_{\tau} \chi_{A_{\tau}}(S) = \gcd_{\tau} m_{A_{\tau}}(S) = S + 2$$

The q -factor, second -2 eigenline and fixed g_3 -cyclic module are not invariants of this normalized internal tilt orbit.

FIGURE 3. The full normalized internal tilt orbit in the fixed envelope. The identification $W \cong \mathcal{R}/\mathbb{C}E$ uses a chosen complement and is not canonical. The six entries of ℓ and the 36 entries of C give 42 parameters. The contrasting representatives leave precisely $S+2$ as the common characteristic and minimal polynomial divisor.

The surviving temporal response is therefore the normalized E -rail:

$$D_{\alpha} A^n E = 2(-2)^n dE \quad (n \geq 0).$$

This is a chain-level boundary response, not the full spectrum of a chosen recurrent matrix.

Remark 4.2. *The dimension seven in this section is an envelope dimension for the present internal tilt calculation. It is not asserted to be invariant under an arbitrary change of partial contraction that changes the recurrent envelope itself.*

Figure 4 displays the all-order response forced by the normalized eigenrelation.

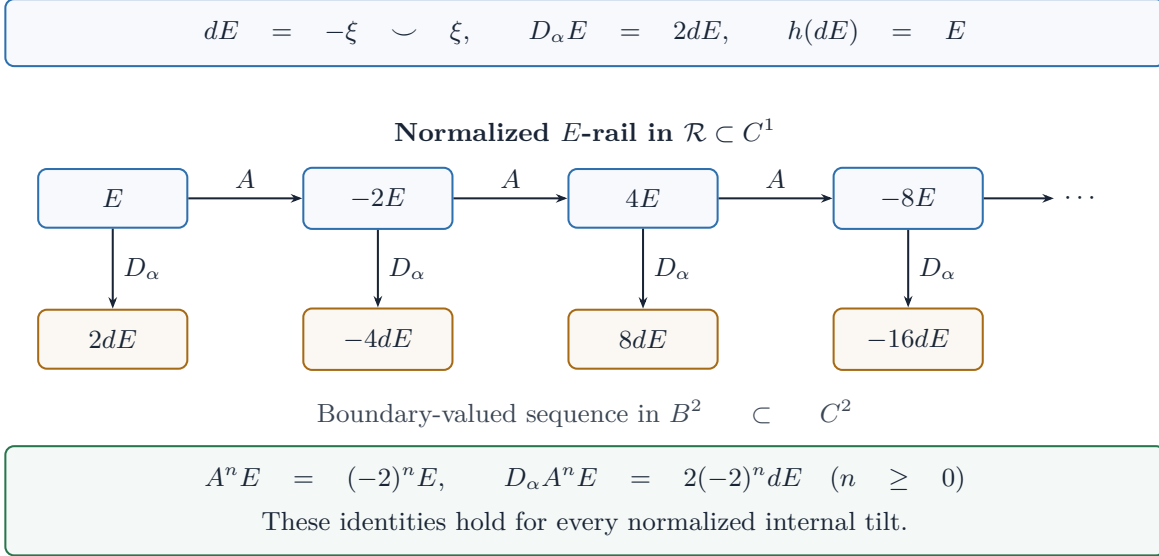


FIGURE 4. The surviving normalized E -rail. The lower row lists boundary-valued outputs of D_α , not an assertion that A acts on C^2 . The formulas hold for every $n \geq 0$ by the eigenrelation and persist throughout the normalized internal tilt orbit.

5. DEPENDENCE OF THE CYCLIC SOURCE ON THE SPLITTING

The fixed low-rail construction has

$$g_3(0) = 2\mathbf{k} - \mathbf{p}.$$

At the first nontrivial source stage, the admissible triangular splitting tilt

$$\mathbf{p}_t = \mathbf{p} + tw, \quad C_{13,t} = C_{13} - tdw,$$

leaves the original source equation and its differential-rank placement unchanged. Taking $w = E$ gives

$$g_{3,t}(0) = g_3(0) - tE,$$

and hence

$$D_\alpha g_{3,t}(0) = D_\alpha g_3(0) - tD_\alpha E.$$

Since $D_\alpha E \neq 0$, the exact g_3 source sequence is not determined by the original equations independently of the contraction. Taking $w = \mathbf{k}$ gives the stronger variation

$$D_\alpha g_{3,t}(0) = D_\alpha g_3(0) - t[\alpha, \mathbf{k}], \quad d[\alpha, \mathbf{k}] \neq 0,$$

so this ambiguity is not merely a boundary ambiguity.

In contrast, the E -rail is forced by the original identities

$$dE = -\xi \smile \xi, \quad [\alpha, E] = -2\xi \smile \xi = 2dE,$$

and the normalized contraction condition $h(dE) = E$. Therefore

$$A(E) = -2E$$

is stable under the normalized internal action.

6. THE PRIMITIVE-FREE RESONANT RESPONSE OBJECT

The E -dependent description is useful for computation but not optimal for comparison. We now remove the primitive choice.

Definition 6.1 (Resonant boundary-response pair). *Let*

$$\beta = \xi \smile \xi, \quad \mathcal{L}_{\text{res}} = \mathbb{C}\beta \subset B^2,$$

and define

$$\mathcal{P}_{\text{res}} = d^{-1}(\mathcal{L}_{\text{res}})/Z^1.$$

The differential induces an isomorphism

$$\bar{d} : \mathcal{P}_{\text{res}} \xrightarrow{\sim} \mathcal{L}_{\text{res}}.$$

Let $T = H^1(C^\bullet)$ with specified cocycle representatives. Since this constrained complex has $C^0 = 0$, we identify it with Z^1 . In the cubic model $T = \mathbb{C}\alpha \oplus \mathbb{C}\xi$ is the two-dimensional Zariski tangent space of $\mathbb{C}[[u, v]]/(v^3)$, not the one-dimensional tangent space of its reduced scheme $\mathbb{C}[[u]]$. Define the linear subspace of response directions

$$T_{\text{resp}} = \{a \in T : [a, Z^1] = 0, [a, d^{-1}(\mathcal{L}_{\text{res}})] \subseteq \mathcal{L}_{\text{res}}\}.$$

For $a \in T_{\text{resp}}$, and only under these conditions, $D_a = [a, -]$ descends to

$$\bar{D}_a : \mathcal{P}_{\text{res}} \longrightarrow \mathcal{L}_{\text{res}}.$$

Definition 6.2 (Response operator and covector). *For $a \in T_{\text{resp}}$, define*

$$\Theta(a) = \bar{D}_a \bar{d}^{-1} \in \text{End}(\mathcal{L}_{\text{res}}).$$

Since $\mathcal{L}_{\text{res}}$ is one-dimensional, there is a unique scalar $\rho_{\text{res}}(a)$ such that

$$\Theta(a) = \rho_{\text{res}}(a) \text{id}.$$

Thus

$$\rho_{\text{res}} \in T_{\text{resp}}^\vee.$$

For the fixed cubic complex of Paper I, $Z^1 = \mathbb{C}\alpha \oplus \mathbb{C}\xi$. The exact cochains satisfy

$$[\alpha, \alpha] = [\alpha, \xi] = 0, \quad [\xi, \xi] = 2\beta \neq 0, \quad [\alpha, E] = 2dE.$$

Since $d^{-1}(\mathcal{L}_{\text{res}}) = \mathbb{C}E + Z^1$, these identities prove $T_{\text{resp}} = \mathbb{C}\alpha$ and $\rho_{\text{res}}(\alpha) = 2$. Thus the response covector is defined on a proper subspace of the two-dimensional tangent space. The extra descent condition cannot be suppressed.

Theorem 6.3 (Maximal descent domain and obstruction). *Put $Z = Z^1$, $V_{\text{res}} = d^{-1}(\mathcal{L}_{\text{res}})$, and let $\pi : C^2 \rightarrow C^2/\mathcal{L}_{\text{res}}$ be the quotient. For the specified tangent representatives, the linear obstruction map*

$$\mathfrak{o} : T \longrightarrow \text{Hom}(Z, C^2) \oplus \text{Hom}(V_{\text{res}}, C^2/\mathcal{L}_{\text{res}}), \quad a \longmapsto (D_a|_Z, \pi D_a|_{V_{\text{res}}})$$

satisfies $T_{\text{resp}} = \ker \mathfrak{o}$. This is the largest linear domain on which the unmodified bracket gives a primitive-independent $\mathcal{L}_{\text{res}}$ -valued response. A strict isomorphism preserving the full tangent representatives and marked boundary line transports this kernel, as well as its response covector.

In the cubic model, if $a = s\alpha + t\xi$ with $t \neq 0$, any linear map $Q : C^2 \rightarrow W$ for which QD_a descends through V_{res}/Z must kill the entire resonant line:

$$Q(\beta) = 0.$$

Thus quotienting the output cannot extend this primitive-free response to such a direction while retaining a nonzero image of the same marked line.

Proof. A map $D_a : V_{\text{res}} \rightarrow C^2$ induces a map from V_{res}/Z to $\mathcal{L}_{\text{res}}$ exactly when it kills Z and has image in $\mathcal{L}_{\text{res}}$. These are precisely the two components of $\mathfrak{o}(a) = 0$; bilinearity proves linearity in a . Preservation of differential and bracket transports both conditions. For the cubic claim,

$$D_a\xi = [s\alpha + t\xi, \xi] = 2t\beta.$$

Since $\xi \in Z$, descent requires $2tQ(\beta) = 0$; over $\mathbb{C}$ this gives the conclusion. Conversely the verified identities show that every $s\alpha$ satisfies both conditions. Therefore $T_{\text{resp}} = \mathbb{C}\alpha$ is exact, not merely a sufficient domain. $\square$

Figure 5 summarizes the primitive quotient, maximal descent domain and transverse obstruction.

7. STRICT RESONANCE-FAITHFUL NATURALITY

Definition 7.1 (Resonance-faithful comparison). *Let two constrained DGLA models carry marked data*

$$(T_{\text{resp}}, \mathcal{L}_{\text{res}}, \mathcal{P}_{\text{res}}, \bar{d}, \bar{D}) \quad \text{and} \quad (T'_{\text{resp}}, \mathcal{L}'_{\text{res}}, \mathcal{P}'_{\text{res}}, \bar{d}', \bar{D}').$$

A comparison consists of isomorphisms

$$\Phi_T : T_{\text{resp}} \xrightarrow{\sim} T'_{\text{resp}}, \quad \Phi_L : \mathcal{L}_{\text{res}} \xrightarrow{\sim} \mathcal{L}'_{\text{res}}, \quad \Phi_P : \mathcal{P}_{\text{res}} \xrightarrow{\sim} \mathcal{P}'_{\text{res}},$$

such that

$$\bar{d}'\Phi_P = \Phi_L\bar{d}, \quad \bar{D}'_{\Phi_T a}\Phi_P = \Phi_L\bar{D}_a.$$

If the comparison comes from a strict DGLA isomorphism, these identities follow from preservation of the differential and bracket together with preservation of the resonance marking.

Theorem 7.2 (Naturality of the marked response). *Under a resonance-faithful comparison,*

$$\boxed{\Theta'(\Phi_T a) = \Phi_L \Theta(a) \Phi_L^{-1}.$$

Consequently,

$$\boxed{\rho'_{\text{res}}(\Phi_T a) = \rho_{\text{res}}(a),}$$

or equivalently

$$\rho'_{\text{res}} = (\Phi_T^{-1})^\vee \rho_{\text{res}}.$$

Proof. Using the two commuting identities gives

$$\begin{aligned} \Theta'(\Phi_T a) &= \bar{D}'_{\Phi_T a}(\bar{d}')^{-1} \\ &= \Phi_L \bar{D}_a \Phi_P^{-1} (\Phi_P \bar{d}^{-1} \Phi_L^{-1}) \\ &= \Phi_L \Theta(a) \Phi_L^{-1}. \end{aligned}$$

The resonant line is one-dimensional, so conjugation leaves its scalar unchanged. The covector formula is the same statement in dual form. $\square$

$$\begin{aligned} \beta &= \xi \smile \xi, & \mathcal{L}_{\text{res}} &= \mathbb{C}\beta \subset B^2, & \mathcal{P}_{\text{res}} &= d^{-1}(\mathcal{L}_{\text{res}})/Z^1 \\ \bar{d} &: \mathcal{P}_{\text{res}} &\xrightarrow{\sim}& \mathcal{L}_{\text{res}} \text{ removes the primitive choice.} \end{aligned}$$

$$\begin{aligned} \text{Full Zariski tangent space: } T &= H^1(C^\bullet) = \mathbb{C}\alpha \oplus \mathbb{C}\xi \\ \text{Response domain: } T_{\text{resp}} &= \mathbb{C}\alpha \\ [\alpha, Z^1] &= 0, & [\xi, \xi] &= 2\beta \neq 0 \end{aligned}$$

$$\begin{aligned} Z &= Z^1, & V_{\text{res}} &= d^{-1}(\mathcal{L}_{\text{res}}), & \pi &: C^2 \rightarrow C^2/\mathcal{L}_{\text{res}} \\ \mathfrak{o} &: T \rightarrow \text{Hom}(Z, C^2) \oplus \text{Hom}(V_{\text{res}}, C^2/\mathcal{L}_{\text{res}}) \\ a &\mapsto (D_a|_Z, \pi D_a|_{V_{\text{res}}}), & T_{\text{resp}} &= \ker \mathfrak{o} \end{aligned}$$

$$\Theta(a) = \bar{D}_a \bar{d}^{-1} = \rho_{\text{res}}(a) \text{id}_{\mathcal{L}_{\text{res}}} \quad (a \in T_{\text{resp}}), \quad \rho_{\text{res}}(\alpha) = 2$$

If $a = s\alpha + t\xi$ with $t \neq 0$, then $D_a\xi = 2t\beta$.
 Descent of QD_a through V_{res}/Z forces $Q(\beta) = 0$.
 Thus output quotienting cannot extend this response to a transverse direction while retaining a nonzero image of the marked resonant line.

FIGURE 5. Primitive-independent response in the cubic model. The full Zariski tangent space is two-dimensional, whereas the exact response domain is $\mathbb{C}\alpha$. An output quotient can permit descent in a transverse direction only by killing the marked line; it cannot retain a nonzero image of that same line.

Remark 7.3. *Rescaling the normal generator changes β and hence the chosen basis of $\mathcal{L}_{\text{res}}$, but the conjugation formula cancels that scalar change. The line and the response operator, rather than a fixed boundary vector, are the intrinsic marked objects.*

The two compatibility identities and their consequence are displayed in Figure 6.

8. ORDINARY QUASI-ISOMORPHISM NO-GO

Theorem 8.1 (Contractible-pair cancellation). *There are strict DGLA quasi-isomorphisms that preserve the formal Maurer–Cartan deformation behavior but erase the marked resonant boundary-response pair.*

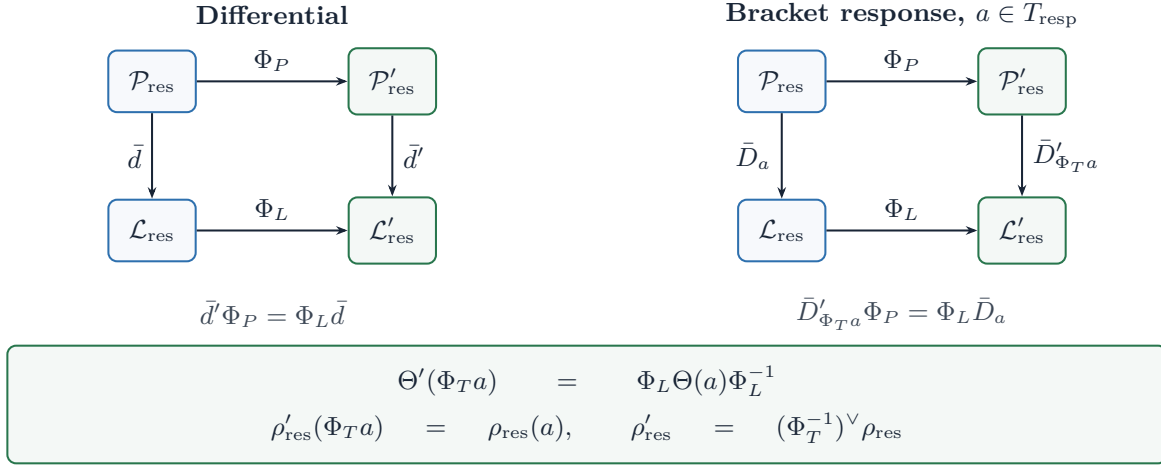
Proof. Let $\mathfrak{g}$ have degree-one generators α, ξ, E and degree-two generator β , with

$$d\alpha = d\xi = 0, \quad dE = \beta,$$

and the only nonzero brackets

$$[\xi, \xi] = 2\beta, \quad [\alpha, E] = 2\beta.$$

Resonance-faithful comparison: isomorphisms Φ_T, Φ_P, Φ_L preserving the marked line and the following two squares.



The scalar response is independent of the basis of the marked line.
Ordinary quasi-isomorphisms need not preserve this line.

FIGURE 6. Resonance-faithful naturality. All three comparison maps are isomorphisms on the specified marked data. The two commuting squares imply conjugacy of the response operators. On the one-dimensional resonant line, this is the stated covector transformation law.

Let $\mathfrak{h}$ be the abelian DGLA on degree-one generators α, ξ , and let

$$p : \mathfrak{g} \longrightarrow \mathfrak{h}$$

kill E and β while fixing α and ξ . All brackets land in the central degree-two line, so the graded Jacobi identity holds. The differential is a derivation because all resulting degree-three terms vanish. The kernel is the acyclic ideal $E \xrightarrow{d} \beta$; hence p is a strict quasi-isomorphism. Both models have two-dimensional degree-one cohomology and vanishing degree-two cohomology.

For $\gamma = u\alpha + v\xi + wE$, the Maurer–Cartan equation in $\mathfrak{g}$ is

$$w + v^2 + 2uw = 0,$$

over a local Artin $\mathbb{C}$ -algebra with u, v, w in its maximal ideal. Since $1 + 2u$ is a unit, $w = -v^2/(1 + 2u)$ is uniquely determined. The assignment is natural in the Artin algebra. There are no degree-zero gauge directions, so this directly identifies the two Maurer–Cartan functors with the same smooth two-parameter germ.

For the response construction, $Z^1 = \mathbb{C}\alpha \oplus \mathbb{C}\xi$ and $T_{\text{resp}} = \mathbb{C}\alpha$: $[\alpha, Z^1] = 0$, whereas $[\xi, \xi] = 2\beta \neq 0$ excludes the ξ -direction. Thus the response covector is defined on this one-dimensional subspace. In $\mathfrak{g}$ one has

$$\mathcal{L}_{\text{res}} = \mathbb{C}\beta, \quad \rho_{\text{res}}(\alpha) = 2,$$

whereas in $\mathfrak{h}$ the product $[\xi, \xi]$ vanishes and

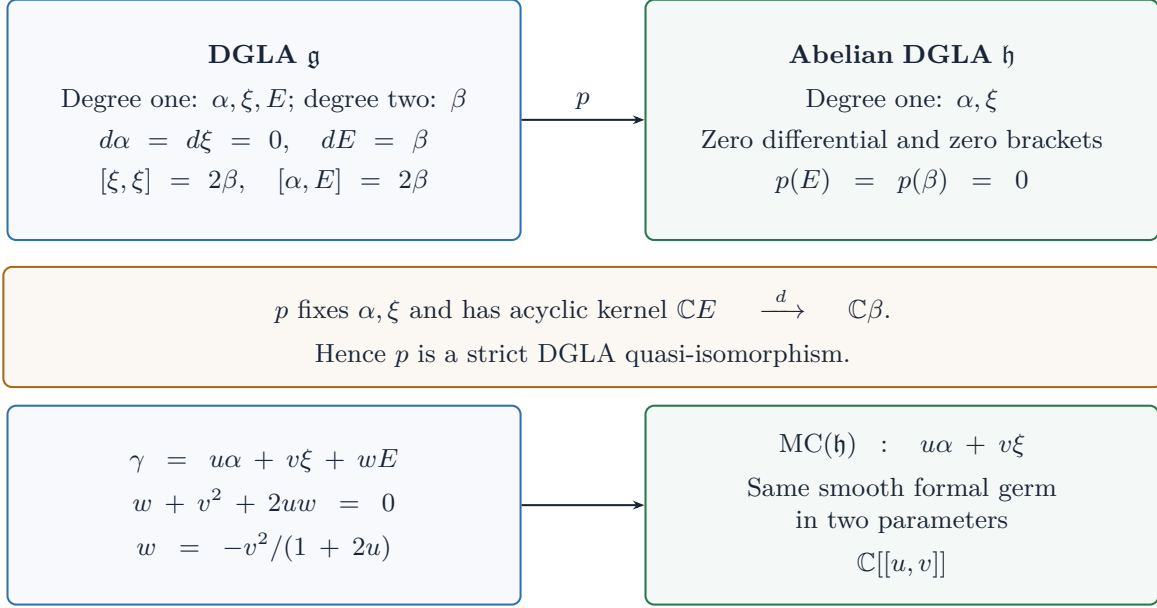
$$\mathcal{L}'_{\text{res}} = 0.$$

Thus the ordinary quasi-isomorphism erases precisely the contractible chain-level datum that the marked theory is designed to retain. $\square$

Corollary 8.2. *The pair $(\mathcal{L}_{\text{res}}, \rho_{\text{res}})$ is not an invariant of the ordinary DGLA derived category or of the associated formal moduli problem alone.*

Figure 7 depicts the independent four-generator example in the preceding proof. Its convention $dE = \beta$ belongs to this example; the earlier cubic cochain satisfies $dE = -\xi \smile \xi$. The smooth germ here is not the nonreduced cubic germ of Paper I.

An independent four-generator counterexample; this is not the cubic model.



Over local Artin algebras, $1 + 2u$ is a unit; there are no degree-zero gauge directions.

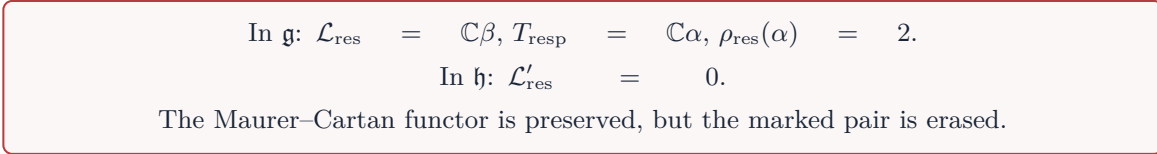


FIGURE 7. Contractible-pair cancellation in an independent DGLA model. Over local Artin algebras, $1 + 2u$ is invertible, so projection identifies the two Maurer–Cartan functors. It simultaneously kills the entire marked resonant line. This is a counterexample to ordinary quasi-isomorphism invariance, not a quasi-isomorphism from the original cubic model to an abelian DGLA.

9. COMPUTATIONAL EVIDENCE AND SCOPE

The finite cubic input is recorded in [6] and is supported by exact rational linear algebra. The locally rerunnable part of the verification family checks:

- the support/incidence model and the 35-dimensional U_{16} controller image;
- the 19-dimensional forced domain and 7-dimensional recurrent image;
- the 42-dimensional internal tilt module and the exact orbit formula;

- the two contrasting characteristic polynomials proving spectral non-invariance.

The bundled supplement reconstructs the Hochschild differentials from the finite algebra action and v0.33 cochain certificate, independently of the unavailable v0.48 JSON and v0.53 helper. It verifies $\dim C_B^1 = 90$, $\text{rank } d_1 = 88$, the two-dimensional cocycle basis, and the source/boundary intersections in Proposition 2.1. It also checks the three g_3 -source identities, the nonzero boundary E -tilt, $d[\alpha, k] \neq 0$, and $d^2k = 0$, together with typed naturality and the cancellation model. Thus these claims have self-contained replacement checks. The all- n E -rail identity follows from $A(E) = -2E$, not from finite samples.

The following questions are intentionally excluded from Paper II:

- (1) a general theorem for arbitrary $m_A(S) = \prod_\lambda (S - \lambda)^{e_\lambda}$;
- (2) contraction-independent classification of the full U_{16} envelope;
- (3) a canonical gauge-fixing that recovers the richer q -spectrum;
- (4) simultaneous deformation of the algebra A and the tensor B .

The response covector is therefore functorial for the specified resonance-faithful comparisons on T_{resp} . The cancellation example proves that its nonzero boundary line cannot be recovered from an ordinary quasi-isomorphism class alone. The cubic polynomial floor and this comparison theorem answer different parts of the same question: what data remain after the permitted changes of splitting and presentation?

Illustrated-edition supplement. The portable source package for this edition includes the seven native vector figure masters and unchanged copies of the five Paper II verification scripts in `evidence/verification/part_II/`, together with the v0.02/v0.03 replacement checks. Run `verify_evidence.py` for the finite mathematical replay and `verify_integrated.py` for source-preservation and PDF checks. The original v0.08 three-paper bundle remains the broader companion archive. The illustrations summarize the manuscript proofs and do not replace them.

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